

Calculus 1: Tutorial serie 3

Exercise 1

1. Calculate in each case, the first three terms of the sequence $(u_n)_n$ defined by the general term u_n , and then give the expression of u_{2n} and u_{n+1}

$$u_n = n^2 + 2n + 3, \quad u_n = \frac{2n - (-1)^{n+1}}{5n + 2}, \quad u_n = \sum_{k=0}^n \frac{(-1)^k}{(k+1)!}, \quad u_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4},$$

$$u_n = \frac{2}{(n-1)(n+2)}, \quad u_n = \frac{2.4.6 \dots (2n)}{1.3.5 \dots (2n-1)}, \quad u_n = \frac{1}{n^2 + 1} + \frac{1}{n^2 + 2} + \dots + \frac{1}{n^2 + n}.$$

2. Prove, using the definition, that the sequence $(u_n)_n$ converges to l :

$$u_n = \frac{n+1}{3n+2}, \quad l = \frac{1}{3}; \quad u_n = \frac{\cos n}{n^2}, \quad l = 0; \quad u_n = \frac{(-1)^{n+1}}{n}, \quad l = 0.$$

Exercise 2 Calculate the limit of the following sequences:

$$u_n = \frac{3n+5}{2n+1}, \quad v_n = \sqrt{5n+1} - \sqrt{5n-1}, \quad t_n = \sum_{k=0}^n \frac{1}{(k+1)(k+2)}, \quad s_n = \frac{\sum_{k=0}^n \alpha^k}{\sum_{k=0}^n \beta^k}, \quad |\alpha| < 1, \quad |\beta| < 1,$$

$$l_n = \frac{3n + (-1)^n}{5n\sqrt{n}}, \quad w_n = (-1)^n + \frac{1}{7n}, \quad w_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}, \quad \theta_n = \sum_{k=1}^n E\left(\frac{k}{2}\right).$$

Exercise 3 Let $(x_n)_n$ be a sequence of positif real numbers, that is decreasing and converges to 0. We set

$$u_n = \sum_{k=1}^n (-1)^k x_k, \quad v_n = u_{2n}, \quad w_n = u_{2n+1}.$$

- 1) Prove that $(v_n)_n$ and $(w_n)_n$ are adjacents.
- 2) Deduce the convergence of $(u_n)_n$.

Exercise 4 Let $\alpha \in]0, \frac{\pi}{2}[$, for all $n \in \mathbb{N}$, we set

$$u_n = 2^n \sin\left(\frac{\alpha}{2^n}\right) \text{ and } v_n = 2^n \tan\left(\frac{\alpha}{2^n}\right).$$

- 1) Prove that $\frac{u_{n+1}}{u_n} = (\cos(\frac{\alpha}{2^{n+1}}))^{-1}$.
- 2) Prove that $(u_n)_n$ and $(v_n)_n$ are adjacents.

Exercise 5

- 1) Prove that $(u_n)_n$ defined by:

$$u_n = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2}, \quad n \in \mathbb{N}^*,$$

is a Cauchy sequence.

- 2) Prove that $(u_n)_n$ defined by:

$$u_n = \frac{1}{\ln 2} + \frac{1}{\ln 3} + \dots + \frac{1}{\ln n}, \quad n \in \mathbb{N}^*,$$

is not a Cauchy sequence.

Exercise 6 Let consider the reccurring sequence $(u_n)_n$ defined by

$$\begin{cases} u_0 = \frac{1}{4} \\ u_{n+1} = \frac{1}{4} + u_n^2. \end{cases}$$

1. Prove that for all $n \in \mathbb{N}$, we have $\frac{1}{4} \leq u_n \leq \frac{1}{2}$.
2. Study the monotonous of $(u_n)_n$
3. Deduce that $(u_n)_n$ is convergent. Calculate its limit.

Exercise 7 Let consider the reccuring sequence $(u_n)_n$ defined by

$$\begin{cases} u_0 = & 1 \\ u_{n+1} = & \frac{1}{2}(u_n + \frac{1}{9u_n}). \end{cases}$$

1. Prove that for all $n \in \mathbb{N}$, we have $u_n > \frac{1}{3}$.
2. Prove that the sequence $(u_n)_n$ is decreasing.
3. Deduce that $(u_n)_n$ is convergent. Calculate its limit.
4. Let $A = \{u_n, n \in \mathbb{N}\}$. Give $\sup(A)$, $\inf(A)$, $\min(A)$ and $\max(A)$ if they exist.

Exercise 8 Let consider the reccuring sequence $(u_n)_n$ defined by

$$\begin{cases} u_0 = & 1 \\ u_{n+1} = & 1 + \frac{2}{1+u_n}. \end{cases}$$

1. Prove that for all $n \in \mathbb{N}$, we have $1 < u_n < 3$.
2. The sequence $(u_n)_n$ is it monotonous?
3. Prove that the subsequences $(u_{2n})_n$ and $(u_{2n+1})_n$ are convergents.
4. Calculate their limit.
5. The sequence $(U_n)_n$ is it convergent? If yes, give its limit.
6. Let $A = \{u_n, n \in \mathbb{N}\}$. Give $\sup(A)$, $\inf(A)$, $\min(A)$ and $\max(A)$ if they exist.

Exercise 9 Let consider the reccuring sequence $(u_n)_n$ defined by

$$u_0 \in \mathbb{R} \text{ and } u_{n+1} = \frac{1}{2} + \frac{1}{1+u_n}.$$

- Study the case $u_0 = 1$.
- Let set now $u_0 = 2$.

1. Prove that for all $n \in \mathbb{N}$, we have $\frac{1}{2} \leq u_n < 2$.
2. Verify that the sequence $(u_n)_n$ is not monotonous.
3. Prove that

$$\forall n \in \mathbb{N}^*, |u_{n+1} - u_n| \leq \frac{4}{9}|u_n - u_{n-1}|,$$

and then that

$$\forall n \in \mathbb{N}^*, |u_{n+1} - u_n| \leq \left(\frac{4}{9}\right)^n |u_1 - u_0|,$$

4. Deduce that $(u_n)_n$ is a Cauchy sequence and give its limit.
5. Give all the values of u_0 that allows that $(u_n)_n$ converges.